

Project Initialization and Planning Phase

Date	11 July 2026
Team ID	xxxxxxx
Project Title	StudyAI – Smart AI Study Companion & Quiz Generator
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report:

Section	Subsection	Content
Project Overview	Objective	The primary objective of StudyAI is to develop an AI-powered study companion that transforms study materials into interactive learning resources, enabling students to generate summaries, flashcards, quizzes, personalized study plans, and performance analytics for effective exam preparation.
Project Overview	Scope	The project provides a complete web-based learning platform that supports uploading PDF, DOCX, TXT files or raw text, AI-powered content generation, adaptive quizzes, study scheduling, weak topic identification, learning analytics, and secure cloud storage using Firebase with Groq AI integration.
Problem Statement	Description	Students often struggle with managing large volumes of study material, creating effective revision notes, preparing quizzes, and organizing study schedules. Traditional learning methods are time-consuming and lack personalization, resulting in inefficient exam preparation.
Problem Statement	Impact	Addressing these challenges

		will improve learning efficiency, reduce study time, increase knowledge retention, provide personalized learning experiences, and help students perform better in examinations through AI-assisted study support.
Proposed Solution	Approach	StudyAI leverages Generative AI using the Groq API (Llama 3.3-70B Versatile), React.js, Python Flask, and Firebase to automatically analyze study materials, generate educational content, identify weak topics, and create adaptive study plans tailored to individual learners.
Proposed Solution	Key Features	• AI-powered study material summarization • Interactive flashcard generation • Adaptive quiz generation (MCQ, True/False, Short Answer) • Personalized 7-day and 30-day study plans • Weak topic identification • Learning analytics dashboard • Progress tracking • Firebase cloud storage • Responsive web application

Resource Requirements

Category	Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Processor and system requirements	Intel Core i5 (10th Gen or above) / AMD Ryzen 5 or equivalent, Optional NVIDIA GTX 1650+ GPU
Hardware	Memory	System RAM	Minimum 8 GB RAM (16 GB Recommended)
Hardware	Storage	Space for application, uploaded files, and logs	Minimum 512 GB SSD
Software	Frameworks	Application frameworks	React.js, Vite, Python Flask
Software	Libraries	Development libraries	Tailwind CSS, Axios, Chart.js, Framer

			Motion, React Markdown, PyPDF2, docx2txt, Groq SDK, Firebase Admin SDK
Software	Development Environment	IDE and development tools	Visual Studio Code, Git, Postman, Python 3.12+, Node.js 22+, npm
Data	Data	Source, storage, and format	User-uploaded PDF, DOCX, TXT files, Raw Text, Firebase Firestore, Local JSON Fallback
Cloud	Hosting & Services	Cloud platforms and AI services	Firebase Firestore, Firebase Authentication, Groq API (Llama 3.3-70B Versatile), Netlify (Frontend), Render (Backend)